Problem 6: Twisted Translations

Word embeddings are numeric representations of words as points in d-dimensional space. That is, a word can be represented using d numbers, where each number represents the coordinate of that number along a particular dimension.

For example, 2-dimensional embeddings are representations of words as points on a plane using a pair of numbers. The first number corresponds to the x-corrdinate of the point and the second number to the y-coordinate of the point.

These word embeddings are very useful representations of words for training machine learning models to perform natural langauge processing tasks. The embedding for a word is computed by looking at the word in a large corpus of text, and looking at what words occur in the context of that word.

Some machine translation models use embeddings of words in the source language to compute the embedding of the word in the target language, and then look up the word corresponding to that embedding. In this problem, we consider translating Hrangkhol to English[7].

[7] Hrangkhol or Hrangkhawl is a Kuki-Chin language spoken by the Hrangkhawl people mainly in Assam and Tripura states in India, with a minority living in Manipur and Mizoram.

Here are some English words, along with their embeddings in two dimensional space:[8]

(a) bull `[2, 2]`
(b) cow `[3, 3]`
(c) mare `[4, 5]`
(d) (female) duck `[3, -1]`
(e) son-in-law `[-5, 1]`
(f) daughter-in-law `[-4, 2]`
(g) grandson `[-6, 3]`

Here are some Hrangkhol words, along with their embeddings in two dimensional space:

(i) vôkchal `[-4, 6]`
(ii) vôkpui `[-6, 8]`
(iii) sakorchal `[-8, 6]`
(iv) arkhong `[4, 2]`
(v) arpui `[2, 4]`
(vi) vatôkhong `[4, 4]`
(vii) vaikikhong `[8, 4]`
(viii) naipa `[-4, -12]`
(ix) nainu `[-6, -10]`
(x) tunu `[-8, -10]`

The English words corresponding to (i-x) in **random** order are:[9]

(A) male parrot
(B) hen
(C) sow
(D) daughter
(E) boar
(F) granddaughter
(G) drake
(H) stallion
(I) son
(J) rooster

Assignment 1: (10 marks)

Match the Hrangkhol words (i-x) to their English translations (A-J).

Assignment 2: (3 marks)

Write down the word embeddings for the Hrangkhol translations of the words (a-g).

Assignment 3: (4 marks)

Write down the word embeddings for the English translations of the words (i-x).

[8]A bull is an adult male cow, and a mare is an adult female horse.
[9]A sow is an adult female pig and a boar is an adult male pig. A stallion is an adult male horse. A drake is an adult male duck. A rooster is an adult male chicken.